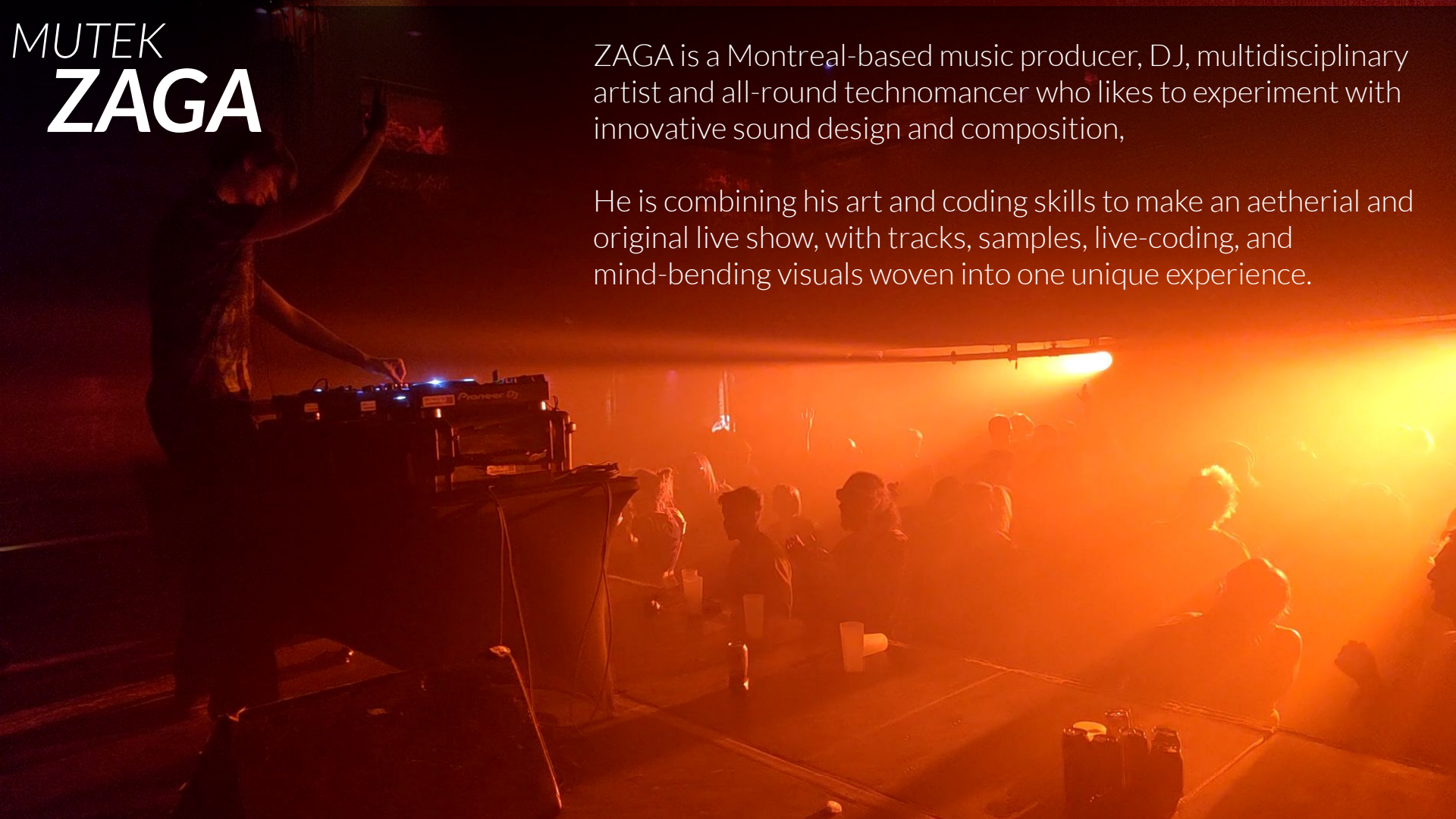


MUTEK
ZAGA

ZAGA is a Montreal-based music producer, DJ, multidisciplinary artist and all-round technomancer who likes to experiment with innovative sound design and composition,

He is combining his art and coding skills to make an aetherial and original live show, with tracks, samples, live-coding, and mind-bending visuals woven into one unique experience.

ZAGA makes electronic music in a wide variety of genres, as well as animations, games and open source audio software.

Selected released tracks can be found on **Bandcamp**, and a further selection of older tracks on **Soundcloud**, An independent personal page is at **ZAGA.MUSIC**.

In his other endeavors, he leads the development of an open source alternative to hearing aids. His influences include a lifelong passion for music and sound, science fiction, and an involvement in hacker culture and electronic music underground.

EXPERIMENTAL ***SOUND DESIGN***

ZAGA uses innovative sound design techniques in all music projects. For example, both of the tracks on the right use a tuning of individual harmonic frequencies, as well as wavetable synthesis, glitch effects, and resynthesis.

Technomancer, above, takes advantage of psychoacoustics to make you hallucinate a melody. We're probably hearing the same one though. Probably. The full song can be experienced interactively at **cubetopia.art/BPM**.

Sequence, below, also tunes atonal harmonics, this time to a chord progression. The video features a real-time particle synth which can be live-synched to the music. These and other songs can be found on **Bandcamp**.

EPIC ***VISUALS***

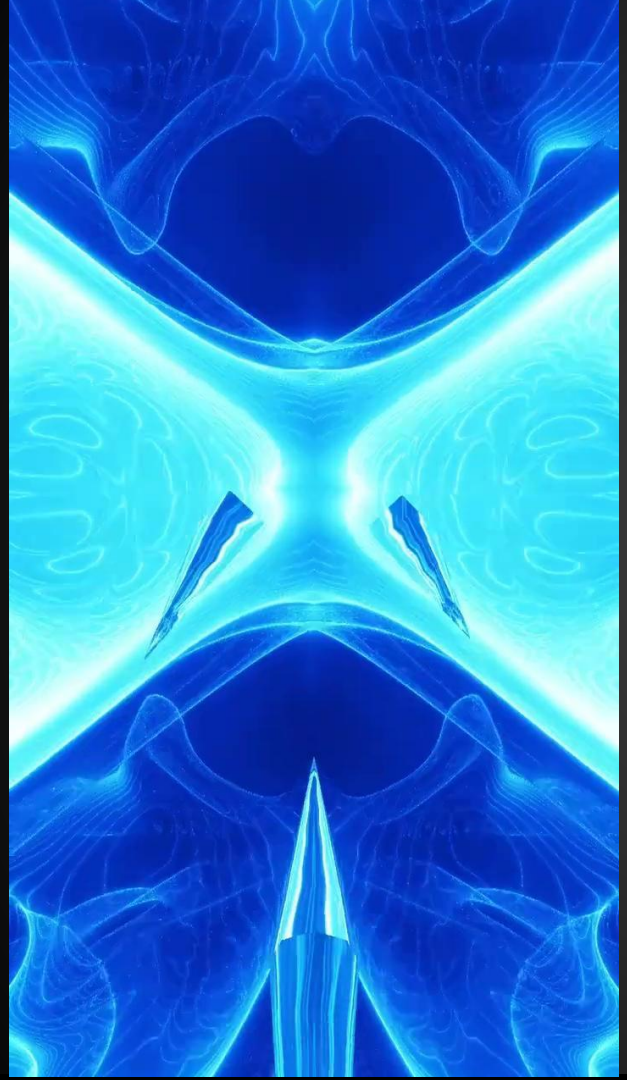
ZAGA makes animations and interactive experiences for his music. The style often features fractal art, retro gaming aesthetic and science fiction.

The rich detail of these makes them appropriate for a venue such as the SAT dome.

The fractal art visuals in particular, need to be pre-rendered as it is very time-consuming and not feasible to generate live.

The video above, *Alien & Health*, is rendered in 360 degrees, and one can use the mouse to look around.

The video below, *Contact*, rendered in 180 degrees could also easily be presented in the dome with the center of the render pointing up - it would give the audience the sensation of being pulled upwards.

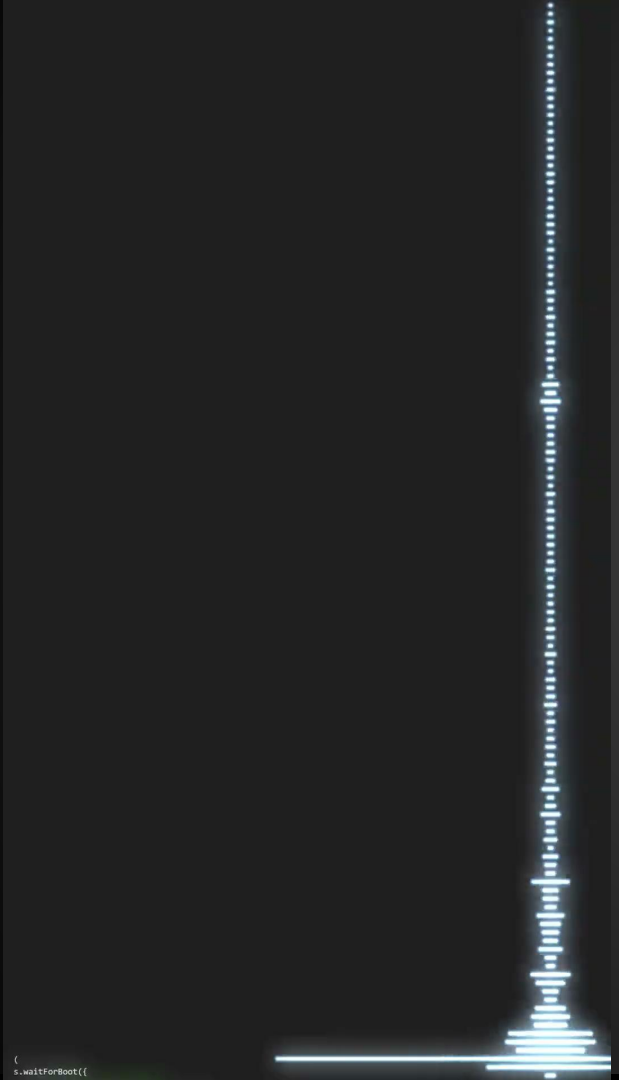


For the parts of the music performance happening live, real time sound-reactive shaders would be the appropriate visual component.

A selection of these can be found in ZAGA's interactive music experiences available at **https://cubetopia.art/BPM.html,** which were made in the Godot game engine and feature many sound-reactive real time visual effects and gameplay mechanics.

An example of it is the *Hyperspace* experience, captured on the left. These would be an easy transfer into a dome setting.

A further showcase of ZAGA visuals is on **YouTube.**



LIVE CODED
MUSIC

ZAGA live codes music, using his experience with sound design and music theory to make it aetherial and original, surprising those who expect music generated from code to be somewhat basic.

The example of ZAGA's algorave melodic techno in the video on the left was written in SuperCollider, and the code reverse-engineers the wave forms of popular synths to achieve this elevated, polished sound. A higher resolution version can be found here: **DOWNLOAD**

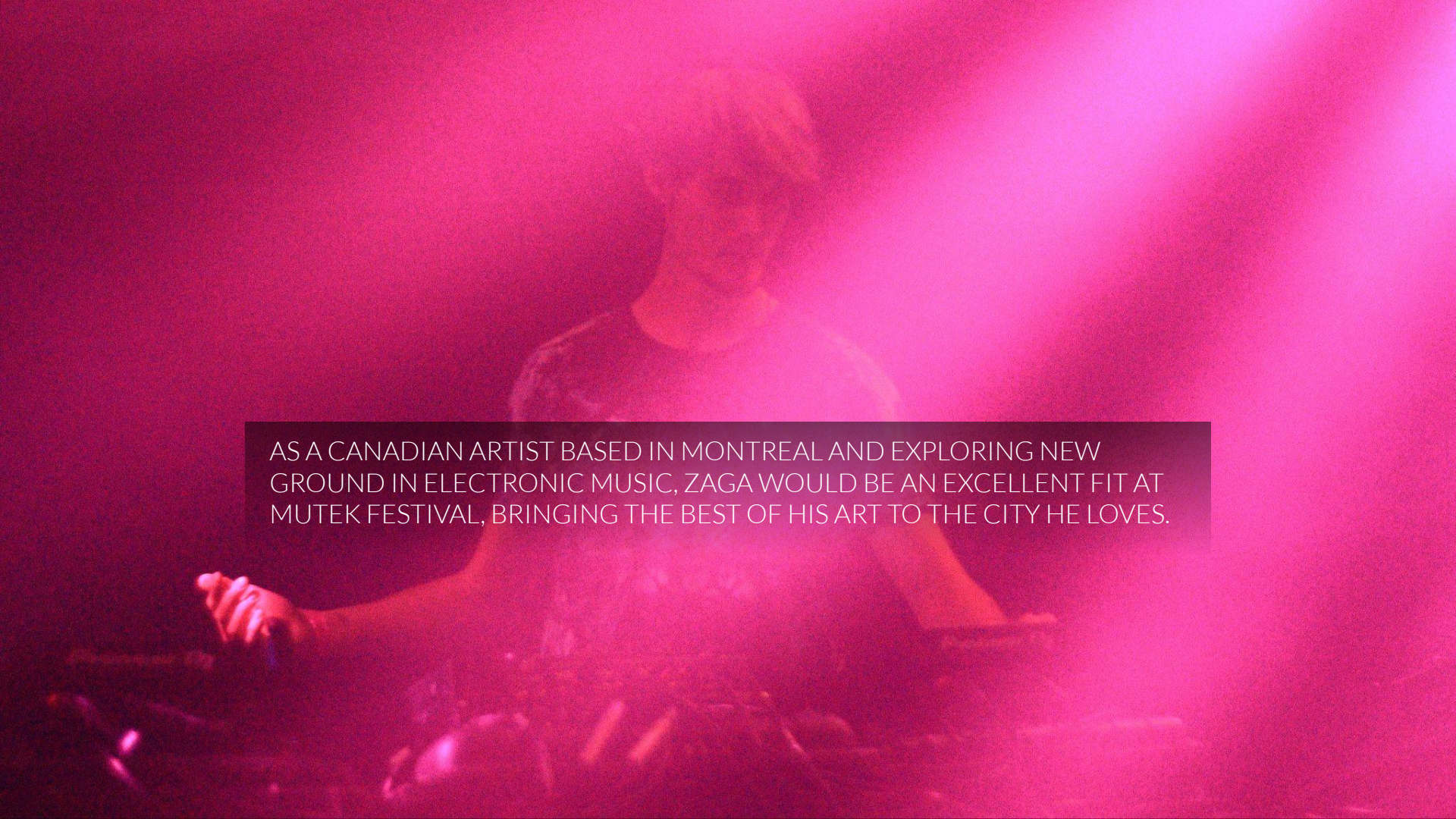
```
// RUMBLE FX: Reverb bass generated from Kick
SynthDef(\rumbleFX, { |inBus, out=0, amp=0.5|
    var input, wet, ducked;
    input = In.ar(inBus, 1);

    // 1. Heavy Reverb
    wet = FreeVerb.ar(input, mix: 1.0, room: 0.9, damp: 0.1);

    // 2. Filter for low end rumble
    wet = LPF.ar(wet, 100);

    // 3. Distortion
    wet = (wet * 3).tanh;

    // 4. Sidechain Ducking (The kick ducks the rumble)
    // Compander: when input (kick) > thresh, compress the wet signal
    ducked = Compander.ar(wet, input,
        thresh: 0.1,
        slopeBelow: 1.0,
        slopeAbove: 0.05, // Crush it when kick hits
        clampTime: 0.002,
        relaxTime: 0.15
    );

    Out.ar(out, Pan2.ar(ducked * amp, 0));
}).add;

// HI-HAT: 909ish
SynthDef(\hat909, { |out=0, amp=0.3, release=0.1|
    var sig, env;
    sig = WhiteNoise.ar;
    sig = HPF.ar(sig, 8000);
    env = EnvGen.ar(Env.perc(0.02, release), doneAction: 2);
    Out.ar(out, Pan2.ar(sig * env * amp, Rand(-0.2, 0.2)));
}).add;

// ACID 303
SynthDef(\acid303, { |out=0, auxBus, send=0, freq=440, gate=1, amp=0.2, cutoff=1000, res=0.2, lag=0.1|
    var sig, env, filt, panned;
    freq = Lag.kr(freq, lag);
    sig = LFSaw.ar(freq);
    env = EnvGen.ar(Env.adsr(0.01, 0.1, 0.6, 0.2), gate, doneAction: 2);

    // Resonant Low Pass
    filt = RLPF.ar(sig, (cutoff * env).clip(50, 18000), res);
    filt = (filt * 2).tanh; // Drive

    panned = Pan2.ar(filt * env, 0);
    Out.ar(out, panned * amp);
    Out.ar(auxBus, panned * send); // Send to FX
}).add;
```

An extensive knowledge of digital synthesis comes in handy here, as just a few lines of code recreate the sound of popular hardware synths such as the Roland TB-303 from frequency information.

In practice, for a live performance some synth definitions are pre-coded to be easily triggered at the beginning of the show, only to be later morphed and modified for a melodic journey.

And it’s not only a journey through sound - but also an eye-opening demonstration of the connection between mathematics and music, that leaves one with a new understanding of the laws of nature.

Live coding can also be used to pass MIDI information to Reaper and trigger synched, pre-synthesized samples and tracks.



AS A CANADIAN ARTIST BASED IN MONTREAL AND EXPLORING NEW GROUND IN ELECTRONIC MUSIC, ZAGA WOULD BE AN EXCELLENT FIT AT MUTEK FESTIVAL, BRINGING THE BEST OF HIS ART TO THE CITY HE LOVES.

THANK YOU FOR YOUR CONSIDERATION!

The audience rioting for an encore after a ZAGA show (video).